Sets (Part 2)

PreCalculus -I

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Membership ($\in$) vs. Subset ($\subseteq$)

Questions:

- Is $\emptyset \subseteq \{1, 2, 3\}$? Yes
- Is $\emptyset \in \{1, 2, 3\}$? No
- Is $\emptyset \subseteq \{\emptyset, 1, 2, 3\}$? Yes
- Is $\emptyset \in \{\emptyset, 1, 2, 3\}$? Yes

More:

- $x \in \{x\}$ Yes
- $\{x\} \subseteq \{x\}$ Yes
- $\{x\} \in \{x, \{x\}\}$ Yes
- $\{x\} \subseteq \{x, \{x\}\}$ Yes
- $\{x\} \in \{x\}$ No

Size of Sets (Cardinality)

Let S be a set.

The cardinality of a set is the number of elements in S, denoted $|S|$.

Examples:

- $A = \{x \mid x \in \mathbb{Z}^+,\ x \text{ is odd},\ x < 10\} = \{1, 3, 5, 7, 9\}, \quad |A| = 5$
- $B = \emptyset, \quad |B| = 0$
- $C = \{\emptyset\}, \quad |C| = 1$
- $\mathbb{R}$ is infinite, so $|\mathbb{R}|$ is infinite.

The Power Set

Let S be a set.

Definition: The power set of S, denoted $\mathcal{P}(S)$, is the set of *all* subsets of S.

Example:

$$\mathcal{P}(\{a, b\}) = \{\emptyset,\ \{a\},\ \{b\},\ \{a, b\}\}$$

Example:

$$\mathcal{P}(\{1, 2, 3\}) = \{\emptyset,\ \{1\},\ \{2\},\ \{3\},\ \{1, 2\},\ \{2, 3\},\ \{1, 3\},\ \{1, 2, 3\}\}$$

What about:

$$\mathcal{P}(\emptyset) = ? \qquad \mathcal{P}(\{\emptyset\}) = ?$$

Cardinality of the Power Set

Assume A is finite.

Examples:

- $A = \{a\}, \quad \mathcal{P}(A) = \{\emptyset, \{a\}\}, \quad |\mathcal{P}(A)| = 2$
- $A = \{a, b\}, \quad \mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}, \quad |\mathcal{P}(A)| = 4$
- $A = \{a, b, c\}, \quad |\mathcal{P}(A)| = 8$

$$\boxed{|\mathcal{P}(A)| = 2^{|A|}}$$

Cartesian Product

Let A and B be sets.

Definition: The Cartesian product of A and B, denoted $A \times B$, is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$.

$$\boxed{A \times B = \{(a, b) \mid a \in A \wedge b \in B\}}$$

Example: $A = \{0, 1, 2\}$, $B = \{a, b\}$.

$$A \times B = \{(0, a), (0, b), (1, a), (1, b), (2, a), (2, b)\}$$
$$B \times A = \{(a, 0), (a, 1), (a, 2), (b, 0), (b, 1), (b, 2)\}$$

So $A \times B \neq B \times A$.

Cardinality of the Cartesian Product

Assume A and B are finite.

Examples:

- $A = \{a\},\ B = \{0\} \quad \Rightarrow \quad A \times B = \{(a, 0)\}, \quad |A \times B| = 1$
- $A = \{a, b\},\ B = \{0\} \quad \Rightarrow \quad A \times B = \{(a, 0), (b, 0)\}, \quad |A \times B| = 2$
- $A = \{a, b\},\ B = \{0, 1\} \quad \Rightarrow \quad |A \times B| = 4$

$$\boxed{|A \times B| = |A| \cdot |B|}$$

Cartesian Product of n Sets

Let $A_1, A_2, \ldots, A_n$ be sets.

Definition: The Cartesian product $A_1 \times A_2 \times \cdots \times A_n$ is the set of all ordered n-tuples $(a_1, a_2, \ldots, a_n)$, where $a_i \in A_i$ for each i.

$$\boxed{A_1 \times \cdots \times A_n = \{(a_1, \ldots, a_n) \mid a_i \in A_i,\ \forall i \in \{1, \ldots, n\}\}}$$

Example: $A = \{a, b\}$, $B = \{1\}$, $C = \{x, y, z\}$.

$$A \times B \times C = \{(a, 1, x),\ (a, 1, y),\ (a, 1, z),\ (b, 1, x),\ (b, 1, y),\ (b, 1, z)\}$$

Cardinality of $A \times B \times C$

Assume A, B, C are finite.

Examples:

- $A = \{a\},\ B = \{0\},\ C = \{x\} \quad \Rightarrow \quad |A \times B \times C| = 1$
- $A = \{a, b\},\ B = \{0\},\ C = \{x\} \quad \Rightarrow \quad |A \times B \times C| = 2$
- $A = \{a, b\},\ B = \{0, 1\},\ C = \{x\} \quad \Rightarrow \quad |A \times B \times C| = 4$

$$\boxed{|A \times B \times C| = |A| \cdot |B| \cdot |C|}$$

Ordered n-tuple

Definition: The ordered n-tuple $(a_1, a_2, \ldots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, ..., and a_n as its nth element.

Example: (a, b) is an ordered 2-tuple (an ordered pair).

Question: Assume $c \neq b$. Are (a, b, c) and (a, c, b) equal?

Answer: No. Although $a = a$, we have $b \neq c$ and $c \neq b$. So the two 3-tuples are not equal.

Order matters in an n-tuple!

Set Notation with Quantifiers

Universal quantifier $\forall x\, P(x)$ over domain S can be written as:

$$\boxed{\forall x \in S\,(P(x)) \quad \equiv \quad \forall x\,(x \in S \rightarrow P(x))}$$

Existential quantifier $\exists x\, P(x)$ over domain S can be written as:

$$\boxed{\exists x \in S\,(P(x)) \quad \equiv \quad \exists x\,(x \in S \wedge P(x))}$$

Example: Universal Quantifier

What does the following statement mean?

$$\forall x \in \mathbb{R}\,(x^2 \geq 0)$$

Solution:

- For every real number x, $x^2 \geq 0$.
- The square of every real number is non-negative.

Example: Existential Quantifier

What does the following statement mean?

$$\exists x \in \mathbb{Z}\,(x^2 = 1)$$

Solution:

- There is an integer x such that $x^2 = 1$.
- There is an integer whose square is 1.

Truth Sets of Predicates

Let P be a predicate and D a domain.

Definition: The truth set of P is the set of elements x in D for which $P(x)$ is true.

$$\boxed{\text{Truth set of } P(x) = \{x \in D \mid P(x)\}}$$

Facts:

- $\forall x\, P(x)$ is true over D iff the truth set of P is the whole set D.
- $\exists x\, P(x)$ is true over D iff the truth set of P is non-empty.

Truth Sets: Examples

Example 1: $P(x)$ is $|x| = 1$, domain $= \mathbb{Z}$.

$$\text{Truth set} = \{-1, 1\}$$

Example 2: $R(x)$ is $|x| = x$, domain $= \mathbb{Z}$.

$$\text{Truth set} = \{x \in \mathbb{Z} \mid x \geq 0\}$$

Example 3: $Q(x)$ is $x^2 = 2$, domain $= \mathbb{Z}$.

$$\text{Truth set} = \emptyset$$

(because $\pm\sqrt{2}$ are not integers).

Summary

Concept	**Key Formula / Idea**
Set	Unordered collection of distinct objects
Membership	$a \in A$ or $a \notin A$
Set builder	$\{x \mid P(x)\}$
Subset	$A \subseteq B \iff \forall x\,(x \in A \to x \in B)$
Proper subset	$A \subset B \iff A \subseteq B \wedge A \neq B$
Equality	$A = B \iff A \subseteq B \wedge B \subseteq A$
Cardinality	$\lvert S \rvert =$ number of elements of S
Power set	$\mathcal{P}(S), \quad \lvert\mathcal{P}(S)\rvert = 2^{\lvert S \rvert}$
Cartesian product	$A \times B = \{(a,b) \mid a \in A,\ b \in B\}, \quad \lvert A \times B \rvert = \lvert A \rvert \lvert B \rvert$
Truth set	$\{x \in D \mid P(x)\}$

Practice Problems

1. Determine whether each statement is true or false.
   1. $2 \in \{1, 2, 3\}$
   2. $5 \subseteq \{1, 2, 3\}$
   3. $\emptyset \subseteq \{1, 2, 3\}$
   4. $\emptyset \in \{\emptyset, 1, 2\}$
2. Write the following set in set-builder notation: $A = \{2, 4, 6, 8, 10\}$.
3. Find $\mathcal{P}(\{x, y\})$ and state $|\mathcal{P}(\{x, y\})|$.
4. Let $A = \{1, 2\}$, $B = \{a, b, c\}$. Find $A \times B$ and $|A \times B|$.
5. Let $P(x)$ be $x^2 = x$, domain $= \mathbb{Z}$. Find the truth set of P.
6. Prove or disprove: If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
7. How many elements does $\mathcal{P}(\emptyset)$ have? What about $\mathcal{P}(\{\emptyset\})$?